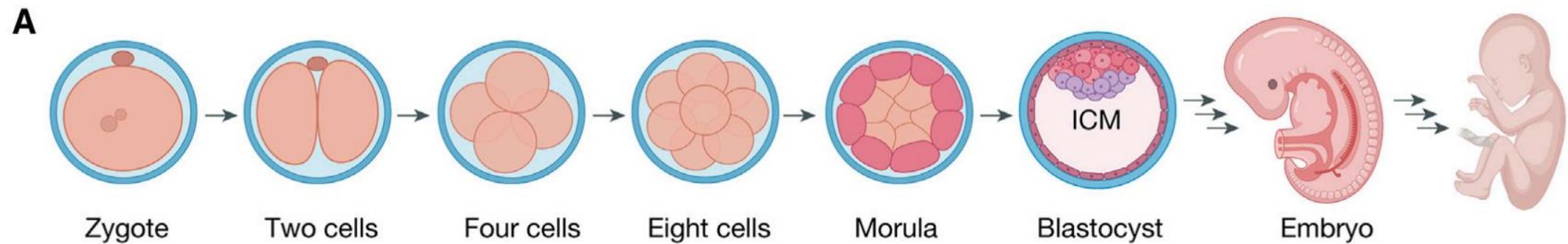


Single-cell phylogenetics for developmental biology

Sophie Seidel

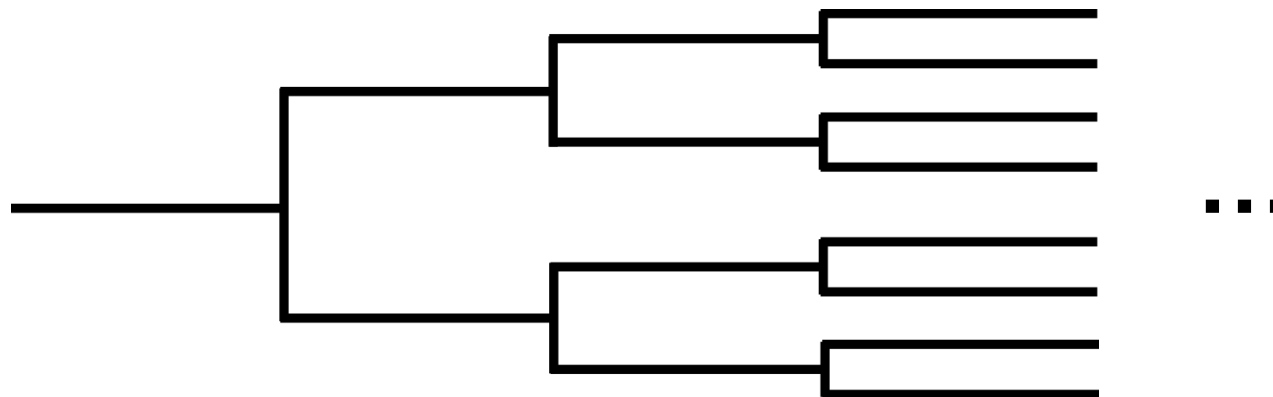
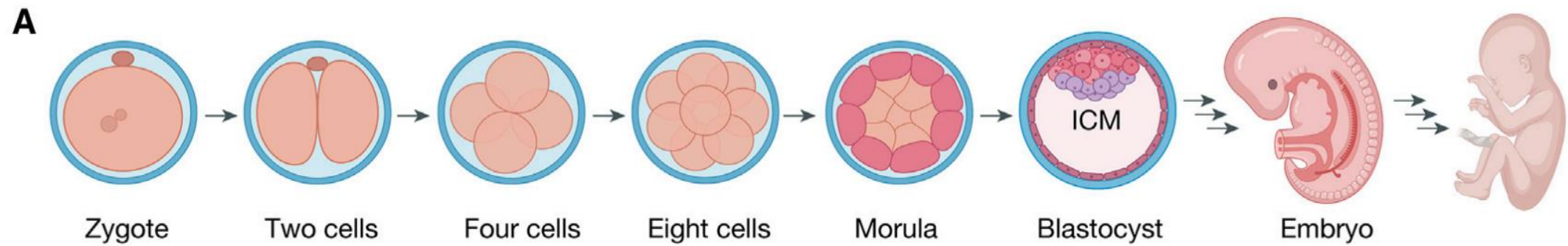
Shendure Lab
Seattle Hub of Synthetic Biology

Understanding development

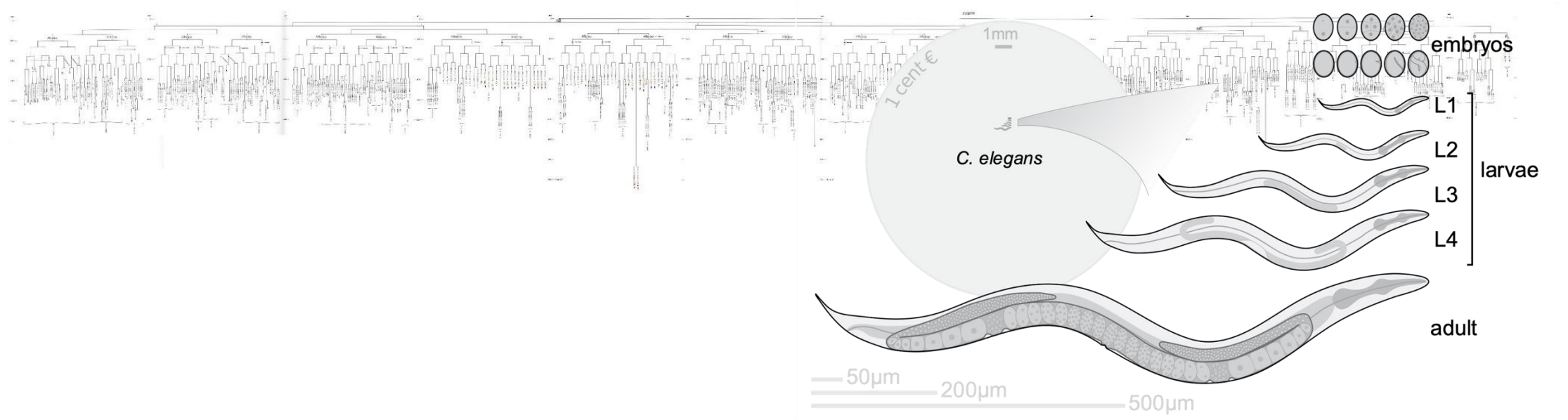


of organisms and tissues

Understanding development



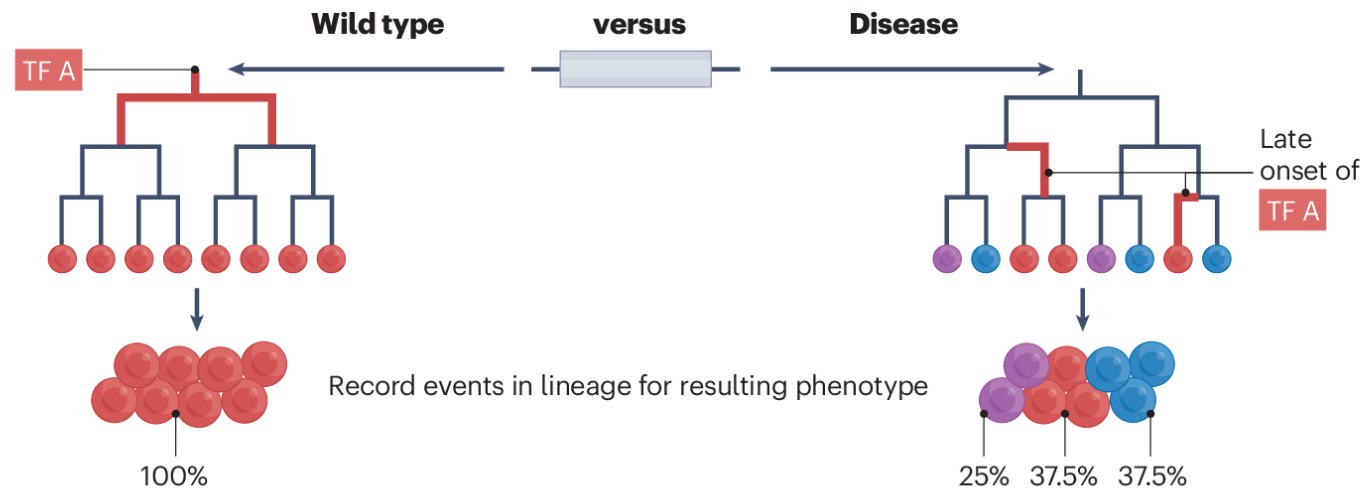
Fundamental research



What is the cell lineage tree in model organisms and humans?

Applied research

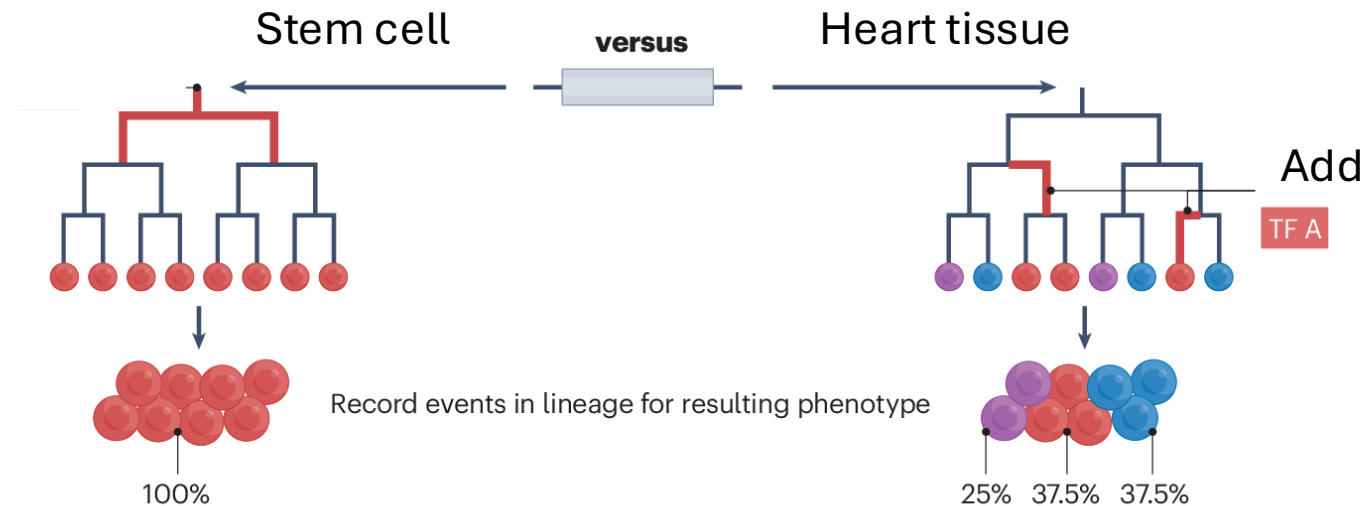
a



How does normal development proceed and what goes wrong in disease?

Engineering biology

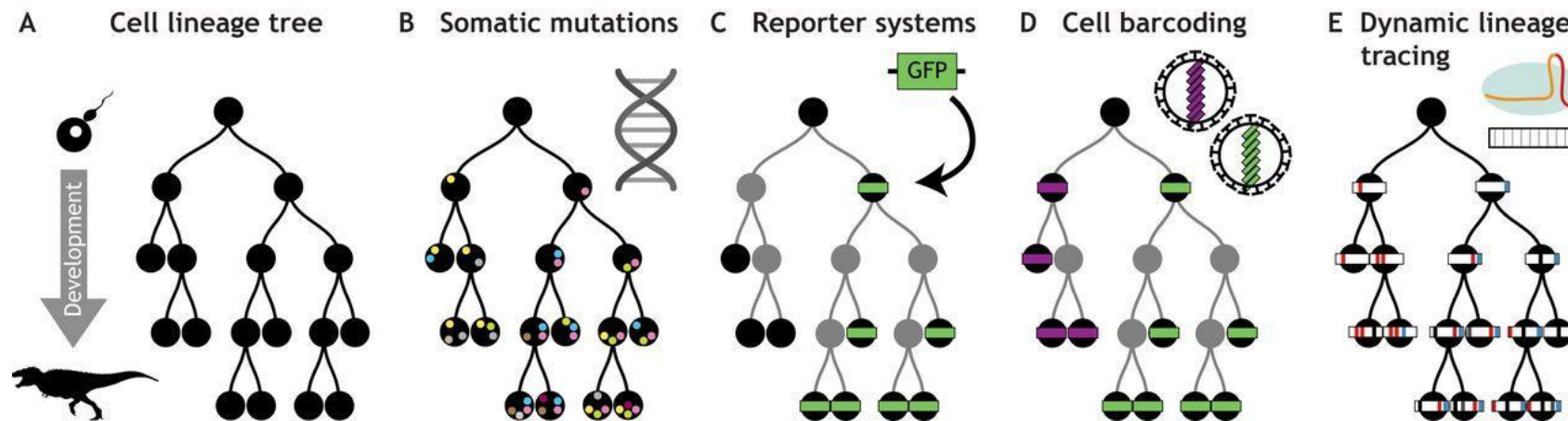
a



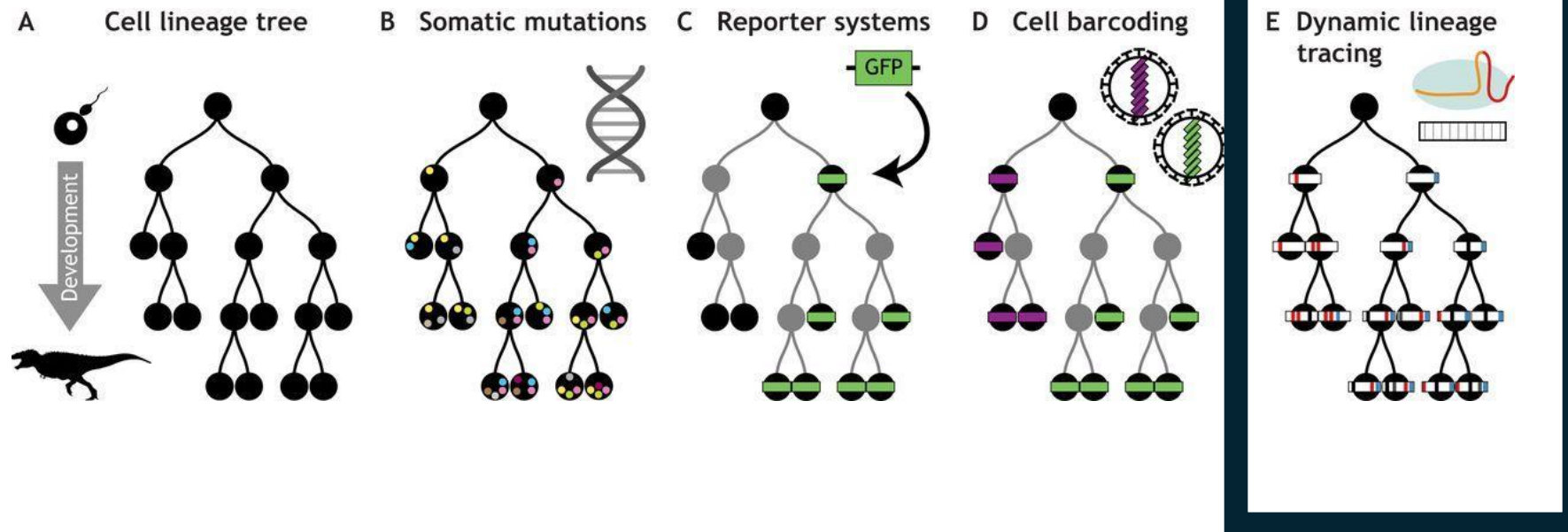
Can we engineer tissues given the knowledge of how cells divide and differentiate?

How to get a cell lineage tree?

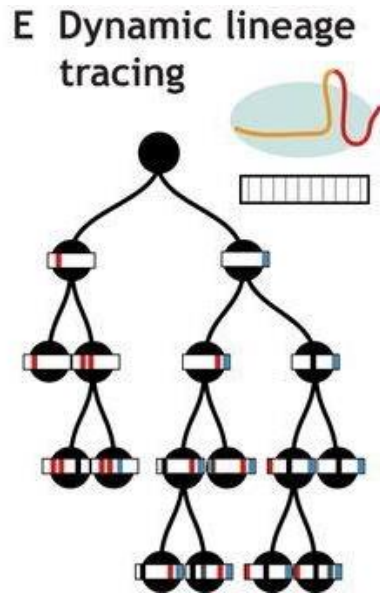
How to get a cell lineage tree?



How to get a cell lineage tree?



How to get a cell lineage tree?

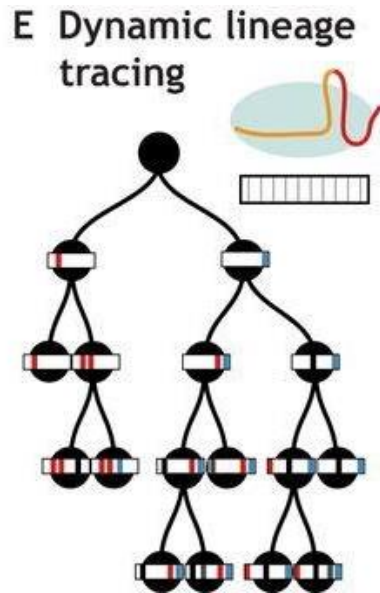


Clone barcode into the genome

Enzyme introduces edits

Edits are inherited over generations

How to get a cell lineage tree?



Clone barcode into the genome

Enzyme introduces edits

Edits are inherited over generations

How can we estimate a phylogeny from such data?

Phylogenetics for dynamic lineage tracing

1. CRISPR Cas9 editing
2. Base editing
3. Prime editing

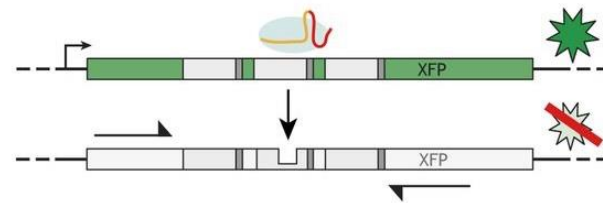
Phylogenetics for dynamic lineage tracing

1. **CRISPR Cas 9 editing**
2. Base editing
3. Prime editing

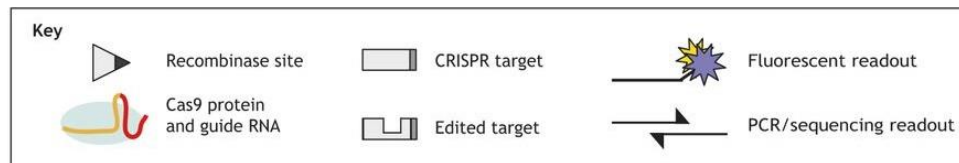
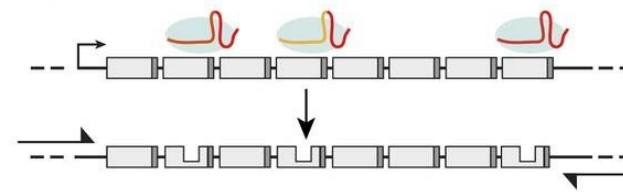
Independent vs Dependent editing

E Sequencing readout of CRISPR barcodes

e.g. ScarTrace and LINNAEUS



e.g. GESTALT and scGESTALT



Standard substitution models

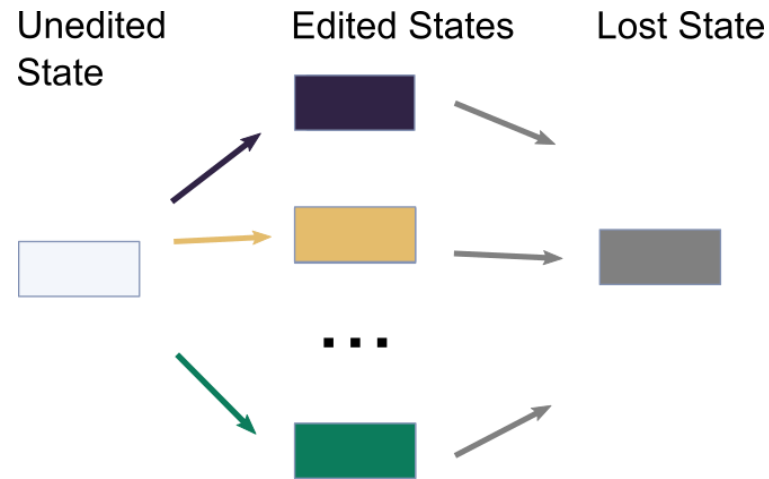
Nucleotides A, C, T, G

Transitions between
nucleotides per site

$$Q = \begin{pmatrix} - & r_{ac} & r_{ag} & r_{at} \\ r_{ca} & - & r_{cg} & r_{ct} \\ r_{ga} & r_{gc} & - & r_{gt} \\ r_{ta} & r_{tc} & r_{tg} & - \end{pmatrix}$$

Equilibrium frequencies at the
root

TiDeTree substitution model



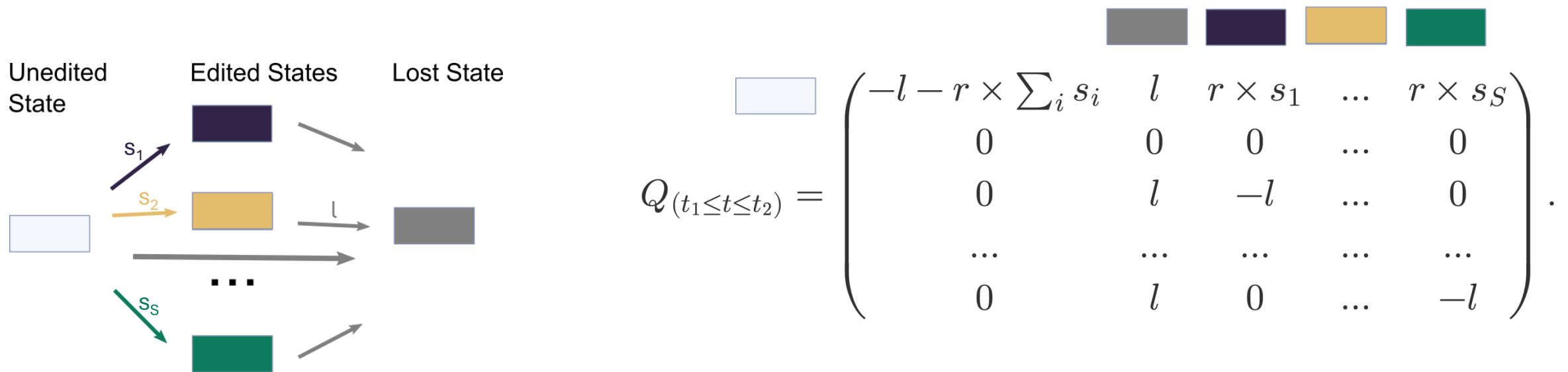
~~Nucleotides A, C, T, G~~

Unedited, edited and lost states

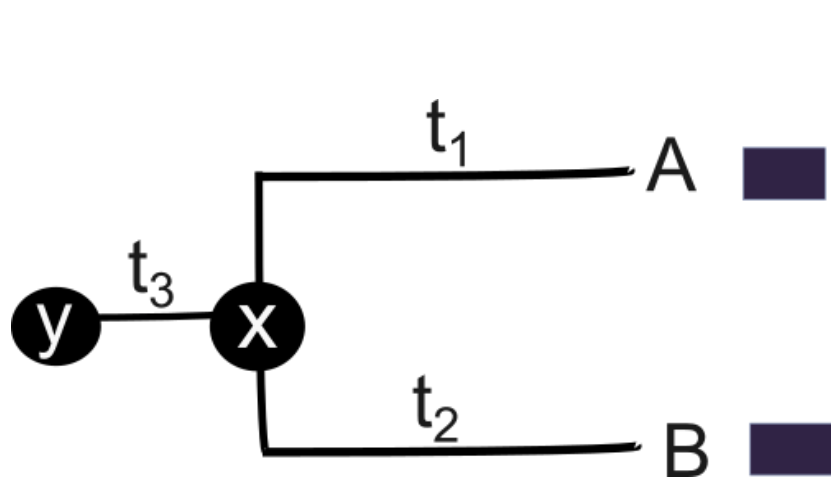
Transitions between
~~nucleotides per site~~ barcode
states

~~Equilibrium frequencies~~ Known
state at the origin

TiDeTree substitution model



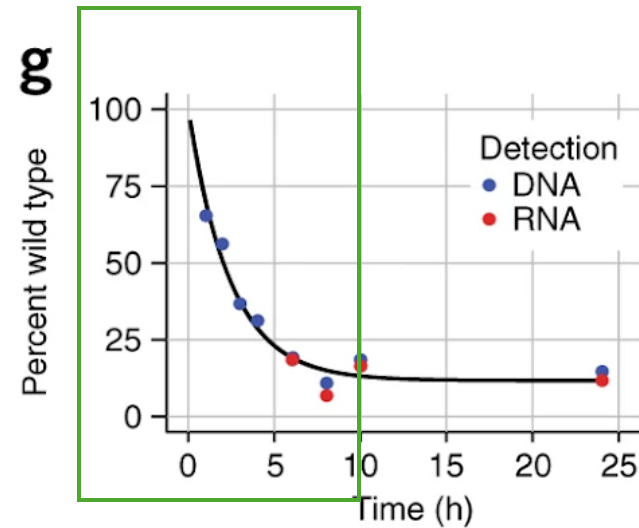
Likelihood computation



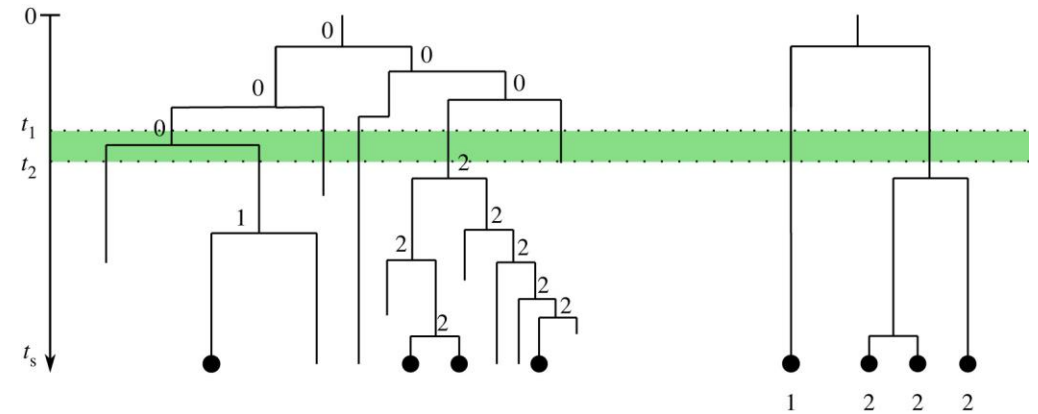
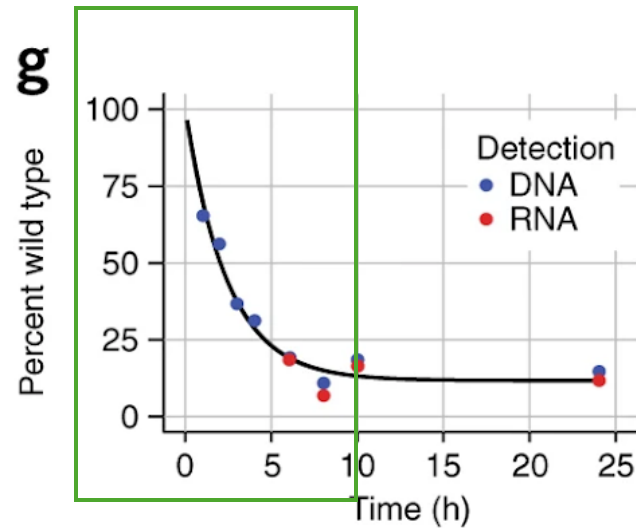
$$Q_{(t_1 \leq t \leq t_2)} = \begin{matrix} \text{light blue box} & \begin{matrix} \text{grey box} & \text{dark blue box} & \text{yellow box} & \text{green box} \end{matrix} \end{matrix} \begin{pmatrix} -l - r \times \sum_i s_i & l & r \times s_1 & \dots & r \times s_S \\ 0 & 0 & 0 & \dots & 0 \\ 0 & l & -l & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & l & 0 & \dots & -l \end{pmatrix}.$$

$$L = \sum_x P_{t_1}(\text{dark blue box} | x) * P_{t_2}(\text{dark blue box} | x) * P_{t_3}(x | y)$$

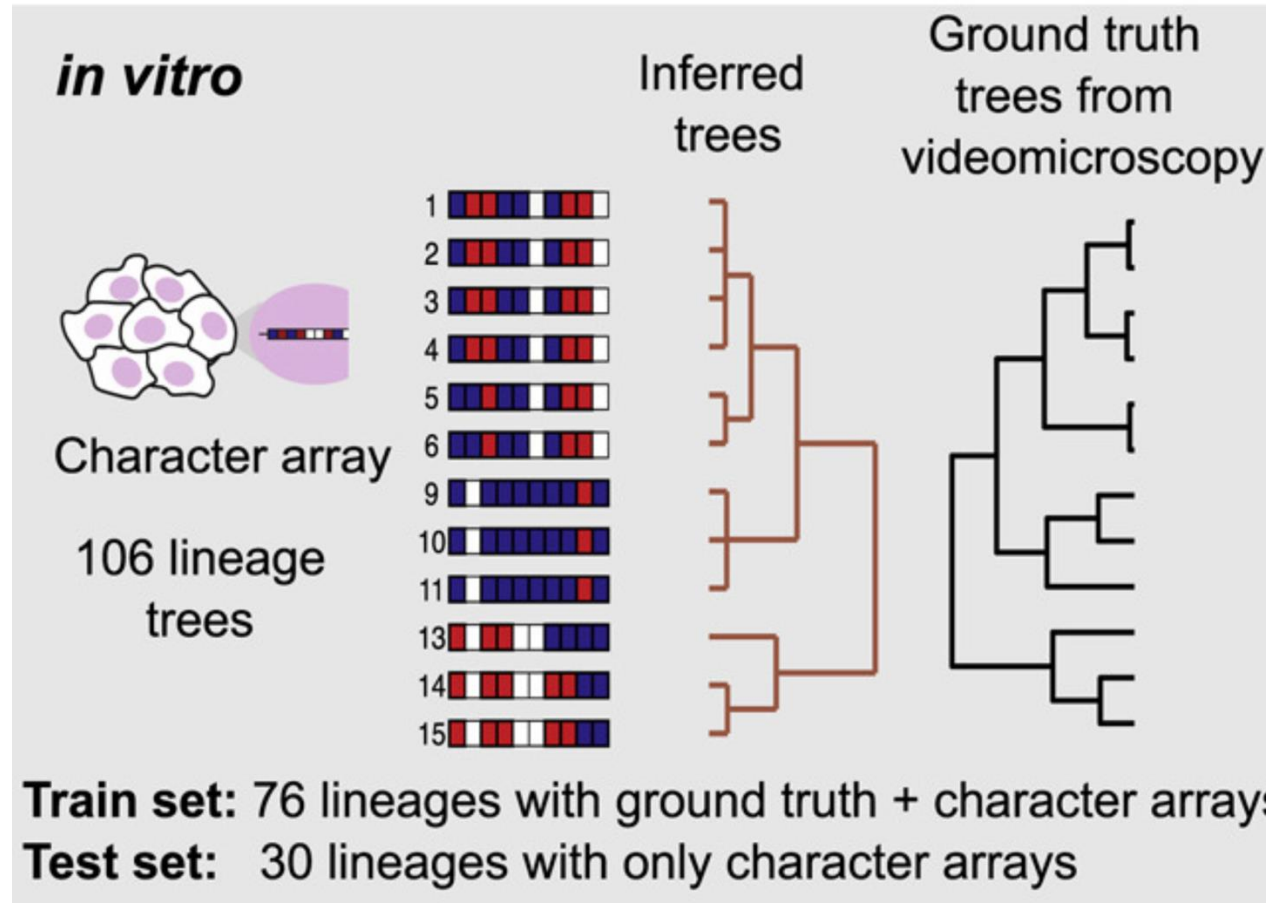
Editing duration is short



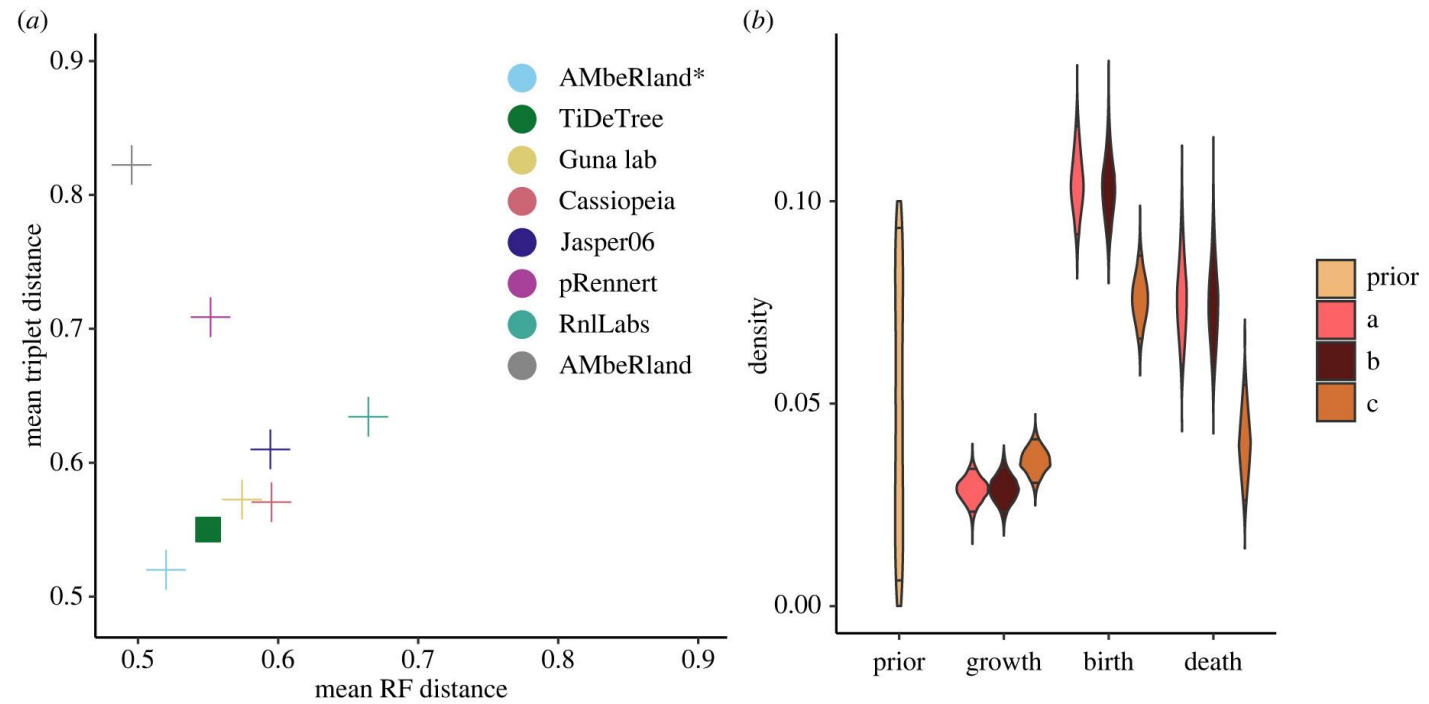
TiDeTree likelihood with piecewise constant editing



Dataset with ground truth trees



TiDeTree



Phylogenetics for dynamic lineage tracing

1. **CRISPR Cas 9 editing**

- 1. Independent sites

- 2. **Dependent sites**

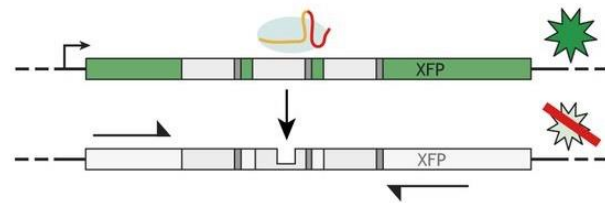
2. Base editing

3. Prime editing

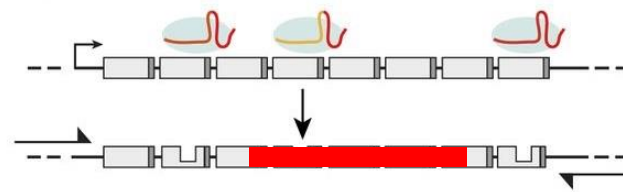
Independent vs Dependent editing

E Sequencing readout of CRISPR barcodes

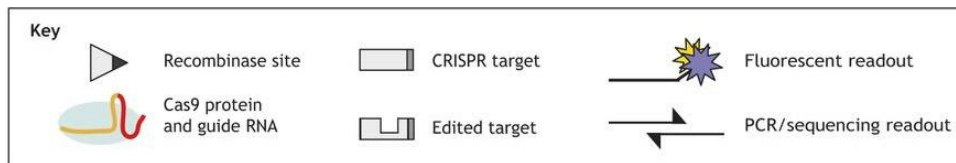
e.g. ScarTrace and LINNAEUS



e.g. GESTALT and scGESTALT

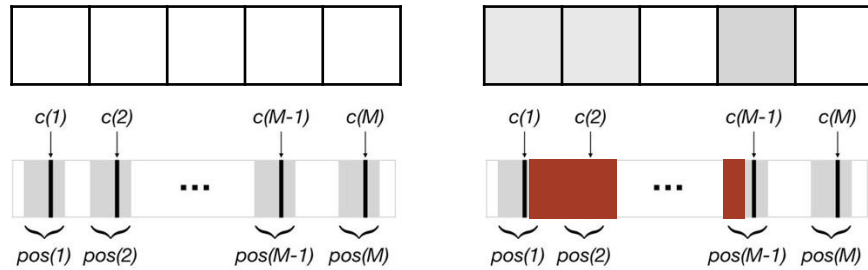


Inter-target deletion introduces dependencies between sites

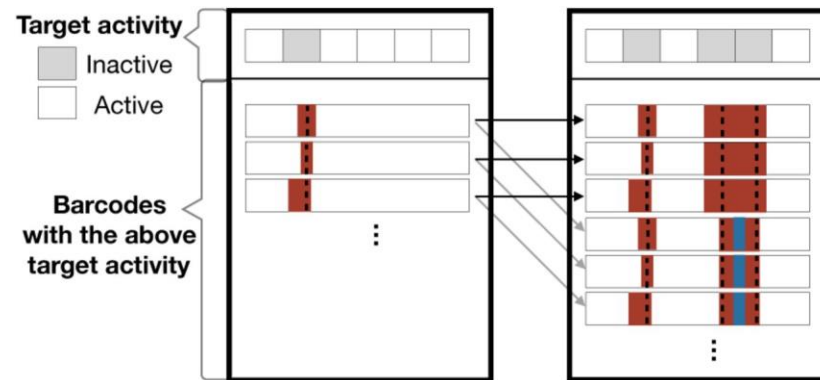


GAPML model

Target activity_{*t*}



Lumped Markov process



Transition rates

$$q(a, \text{Apply}(a, d)) := \lim_{\Delta \rightarrow 0} \frac{\Pr(X(\Delta) = \text{Apply}(a, d) \mid X(0) = a)}{\Delta}$$

$$= h(\tau, \text{TargStat}(a)) \Pr(d \mid \tau).$$

Clock rate

Indel probability

Available in BEAST 2

README

GPL-3.0 license

GABI - GESTALT Analysis using Bayesian Inference in BEAST2:

In this repository, we develop a BEAST 2 package to estimate time-scaled lineage trees and cell population dynamics from GESTALT data, implementing the phylogenetic likelihood in [GAPML](#).

Building from source

The default target in the provided [Apache ANT](#) build script can be used to build the BEAST 2 package from scratch. The package archive will be left in the `dist/` directory.

License

This software is free (as in freedom). With the exception of the libraries on which it depends, it is made available under the terms of the GNU General Public Licence version 3, which is contained in this directory in the file named COPYING.

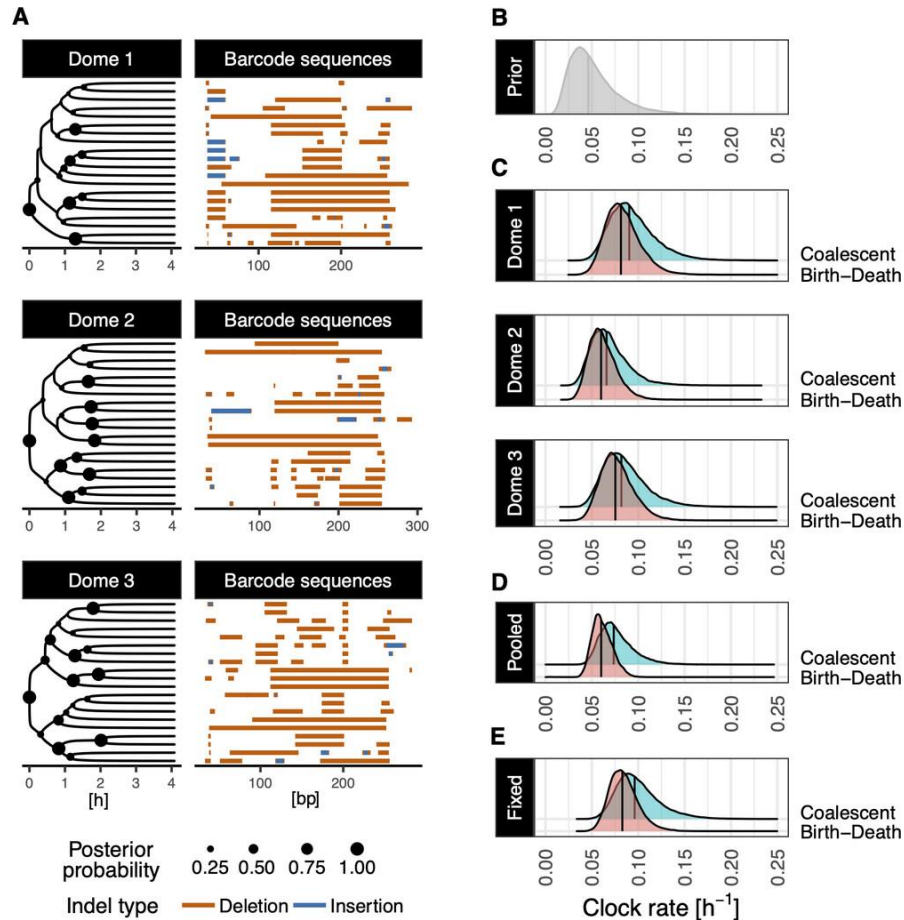
Contributors 2

 **seidels** Sophie Seidel

 **azwaans** Antoine Zwaans

Languages

Java 100.0%



Major limitation

Short editing duration / limited #symbols

→ limited signal in trees

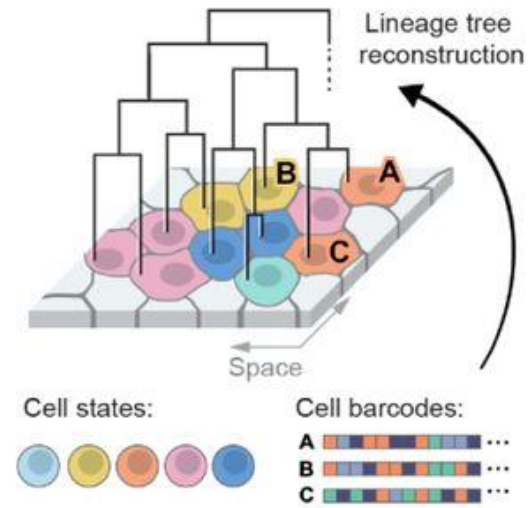
Emergence of prime and base editing technologies

Phylogenetics for dynamic lineage tracing

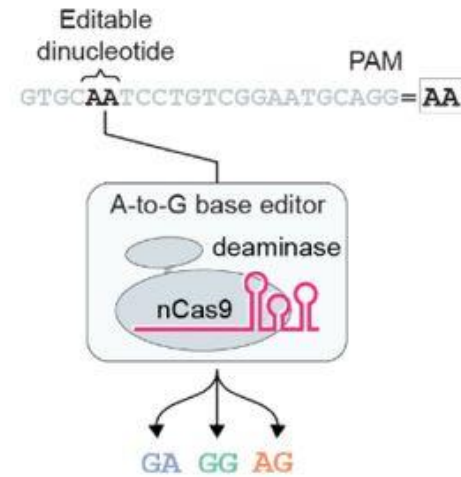
1. Independent sites barcode editing
2. Dependent sites barcode editing
3. **Base editing**
4. Prime editing

Base editing

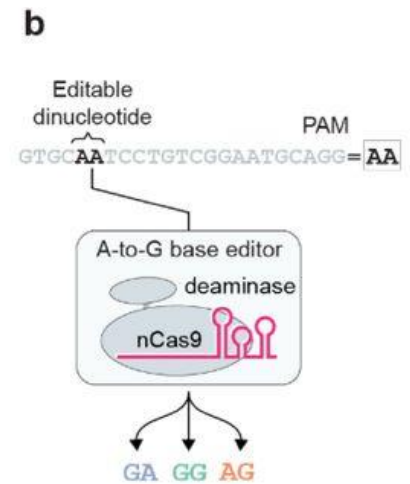
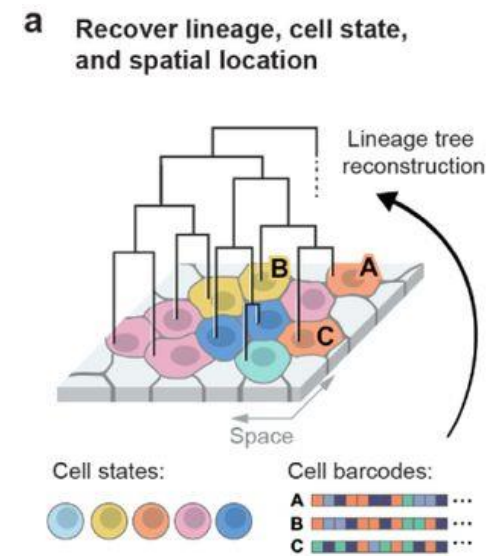
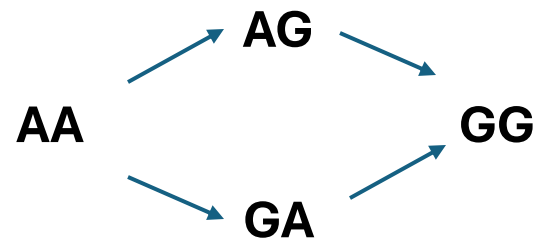
a Recover lineage, cell state, and spatial location



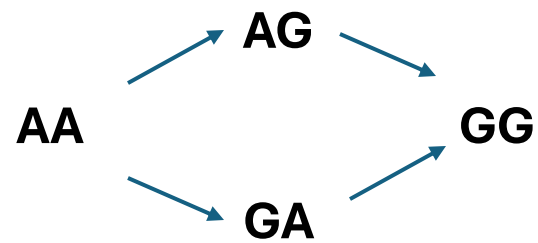
b



Base editing



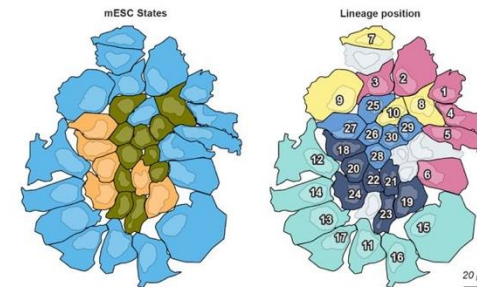
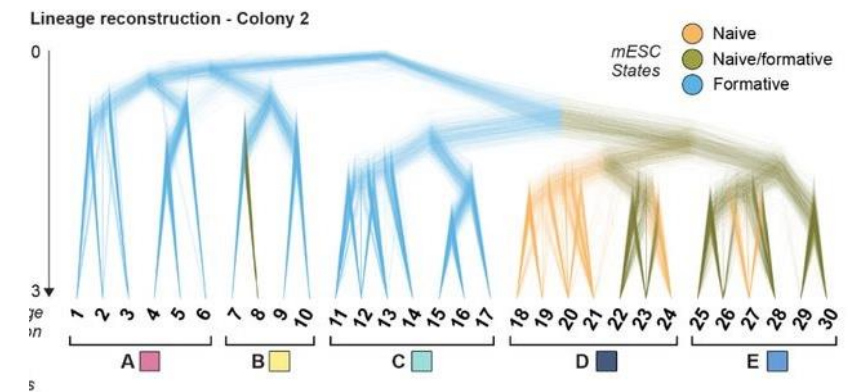
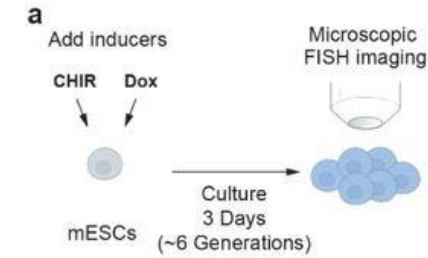
Base editing



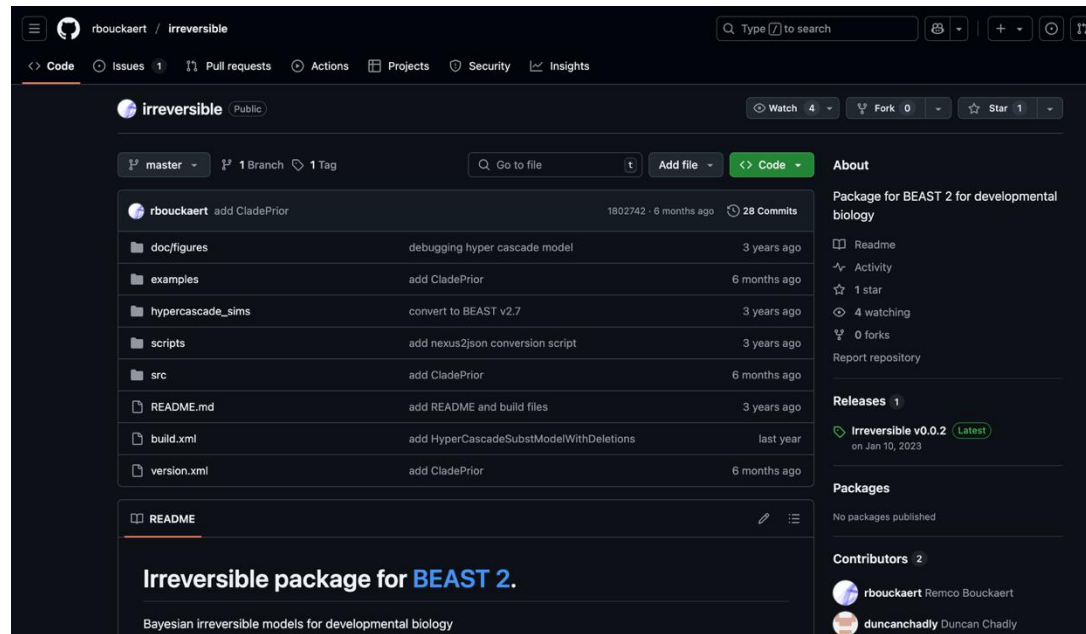
$$Q = \begin{array}{ccccc} & - & r_{AA \rightarrow AG} & r_{AA \rightarrow GA} & 0 \\ 0 & & - & 0 & r_{AG \rightarrow GG} \\ 0 & & 0 & - & r_{GA \rightarrow GG} \\ 0 & & 0 & 0 & - \end{array}$$

Base editing

- 7 colonies
- Shared model parameters across colonies
- Estimated cell state transitions



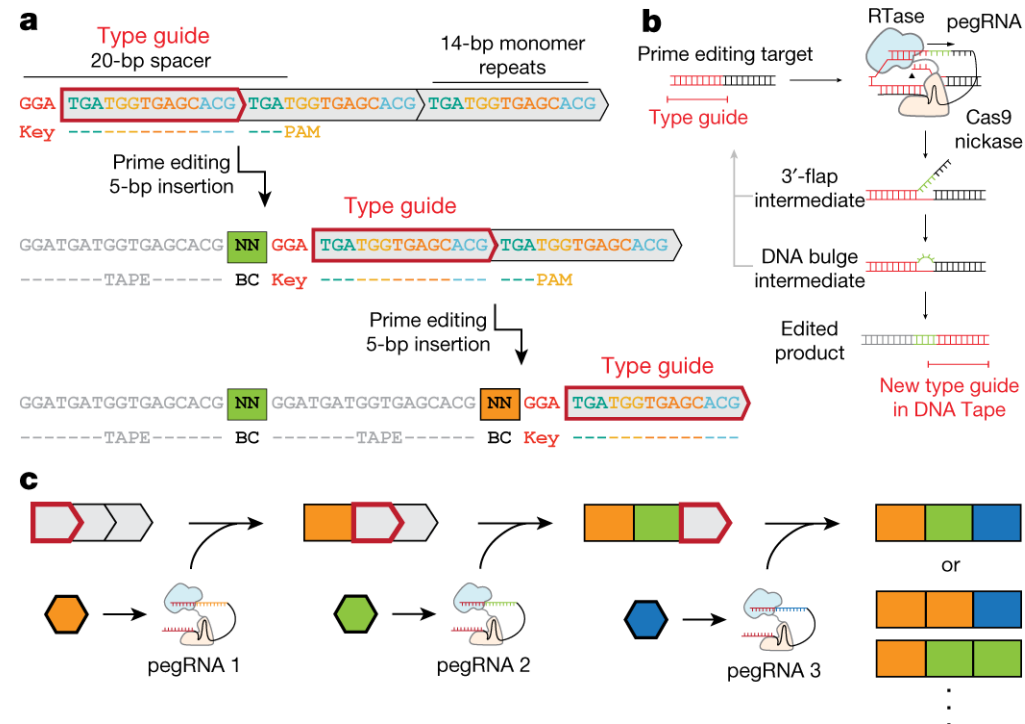
Available in BEAST 2



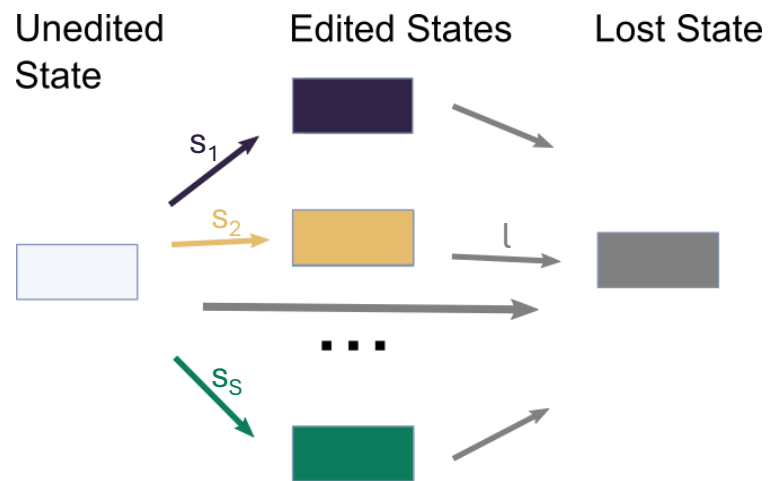
Phylogenetics for dynamic lineage tracing

1. Independent sites barcode editing
2. Dependent sites barcode editing
3. Base editing
4. **Prime editing**

Prime editing



TiDeTree Substitution Model

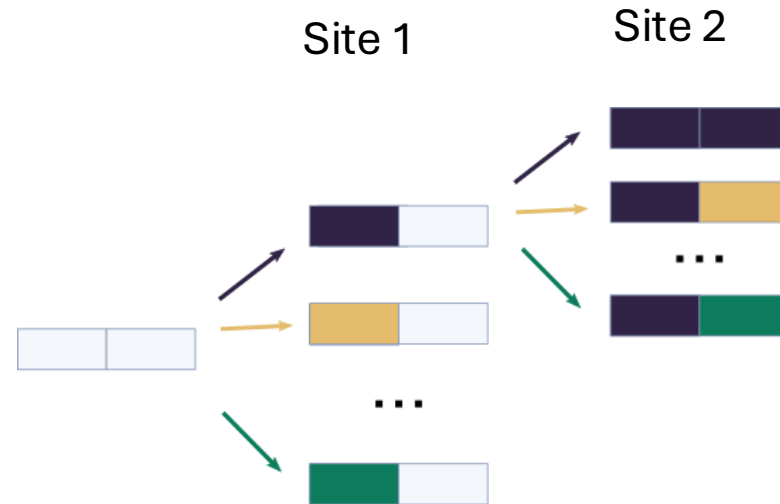


SciPhy substitution model – clock rate



$$P(j; r, \Delta t) = \begin{cases} \frac{(r \Delta t)^j}{j!} e^{-(r \Delta t)} & \text{if unsaturated} \\ 1 - \sum_{k=0}^{j-1} \frac{(r \Delta t)^k}{k!} e^{-(r \Delta t)} & \text{otherwise} \end{cases}$$

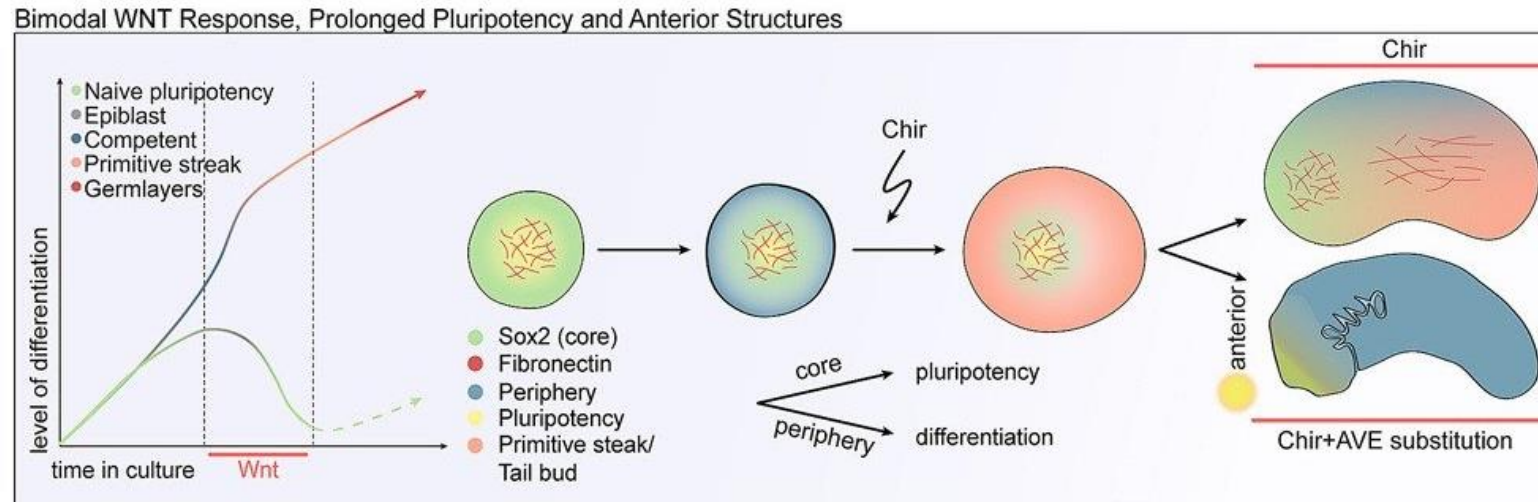
SciPhy substitution model – edit probs



Edit outcome probabilities

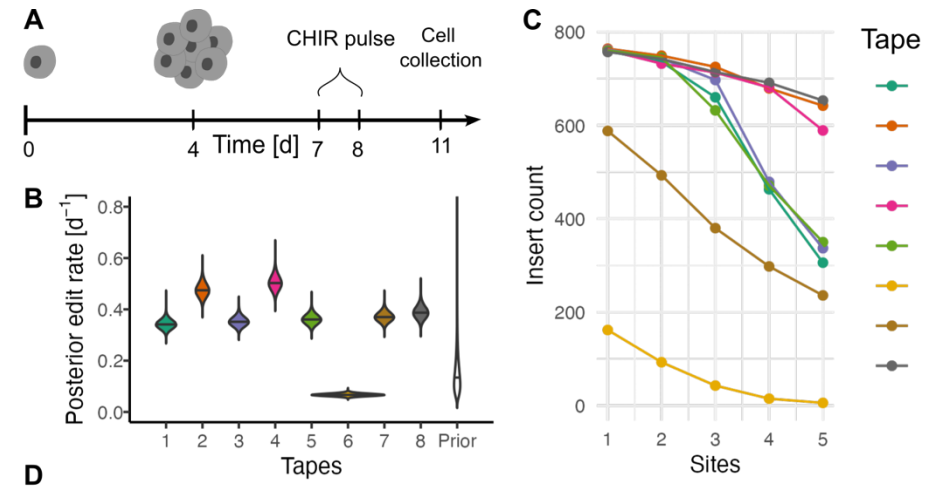
Monoclonal gastruloid

a model of early embryonic development



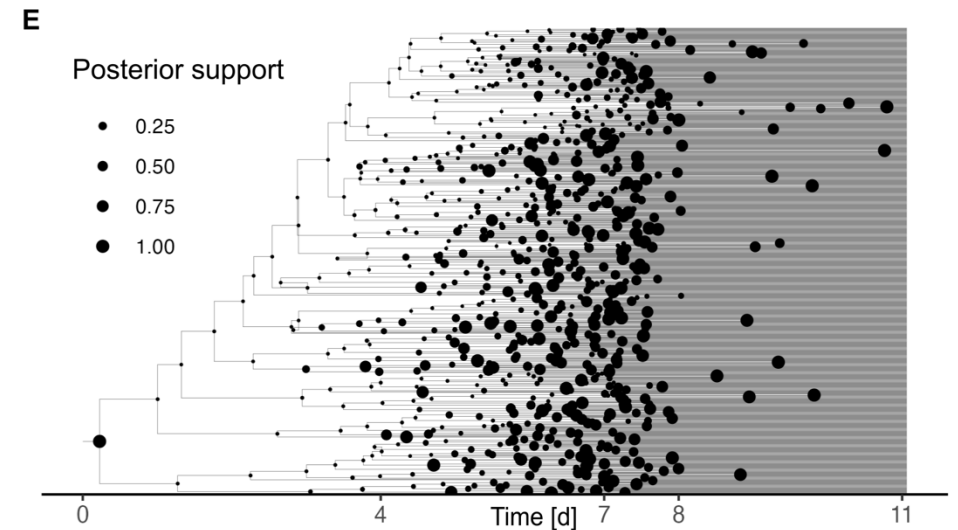
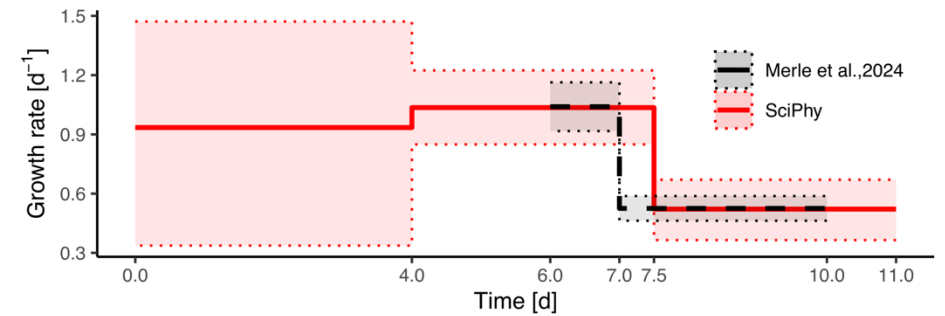
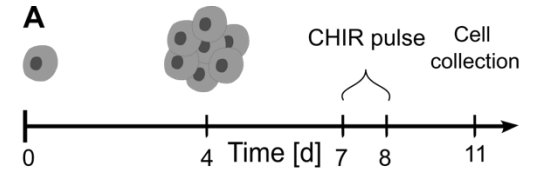
Analysis of gastruloid data

- Experiment with temporal information
- Estimate clock rate per barcode

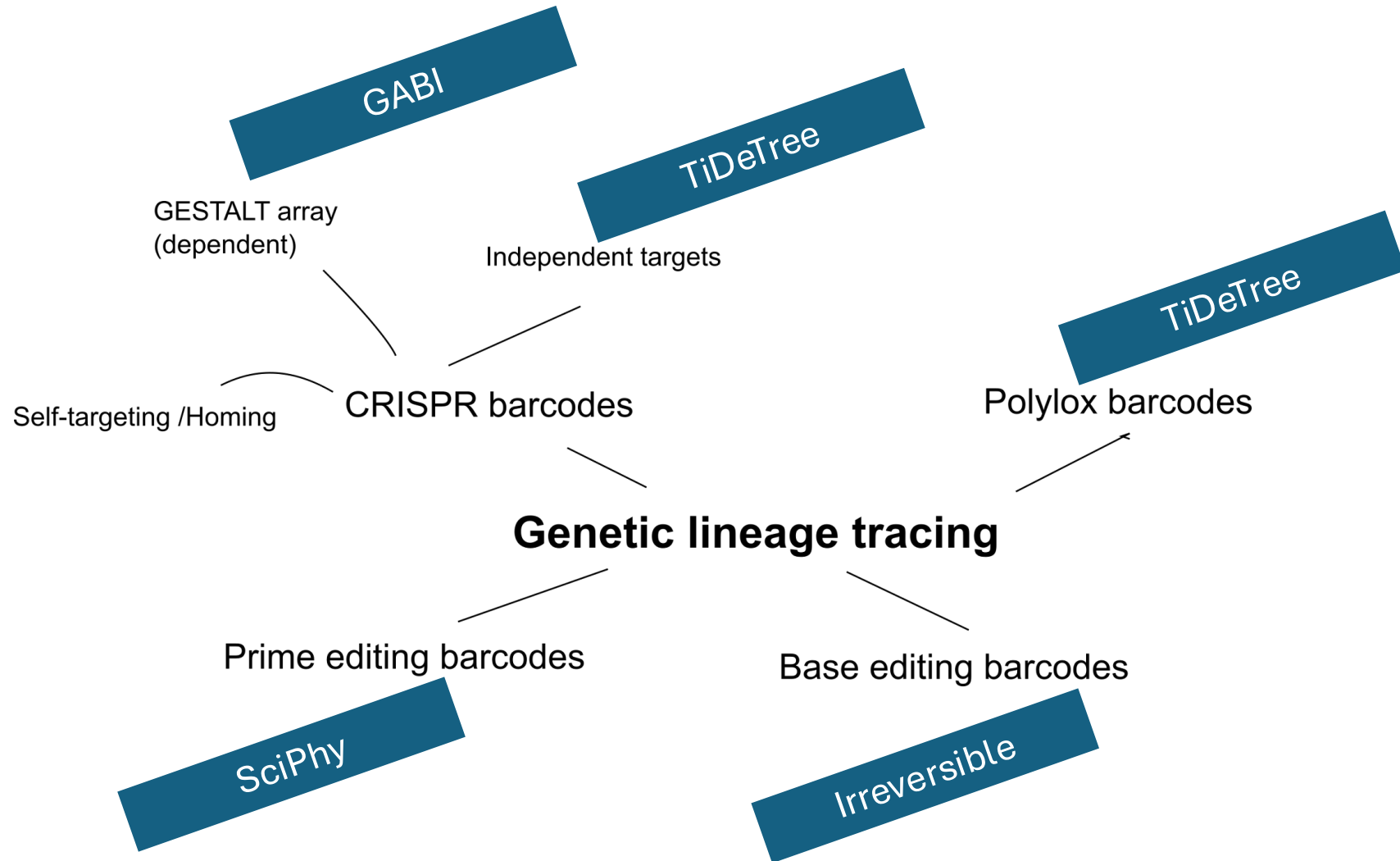


Analysis of gastruloid data

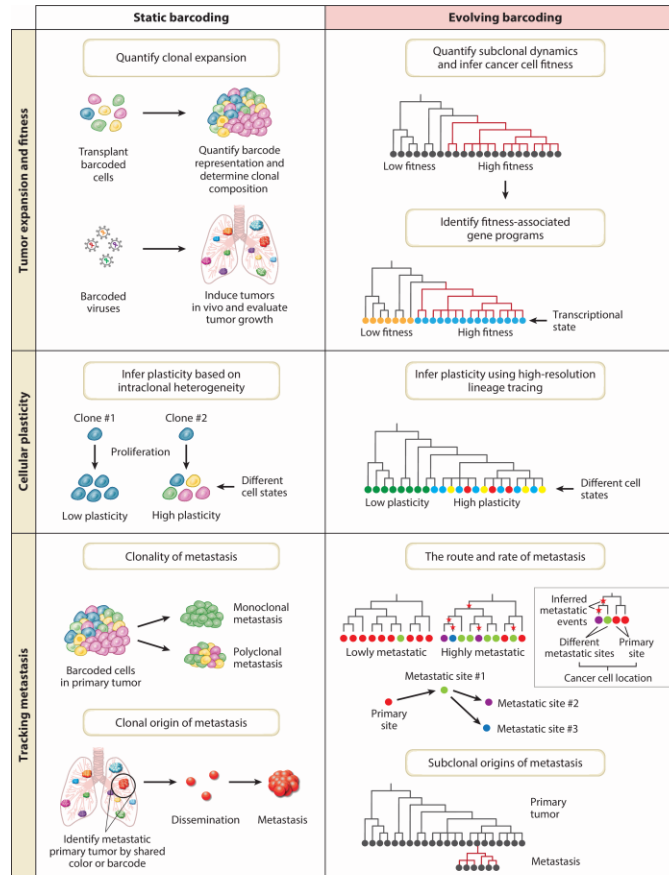
- Experiment with temporal information
- Estimate clock rate per barcode
- Estimate birth-death skyline
- Identified slowdown



BEAST 2 packages for lineage tracing



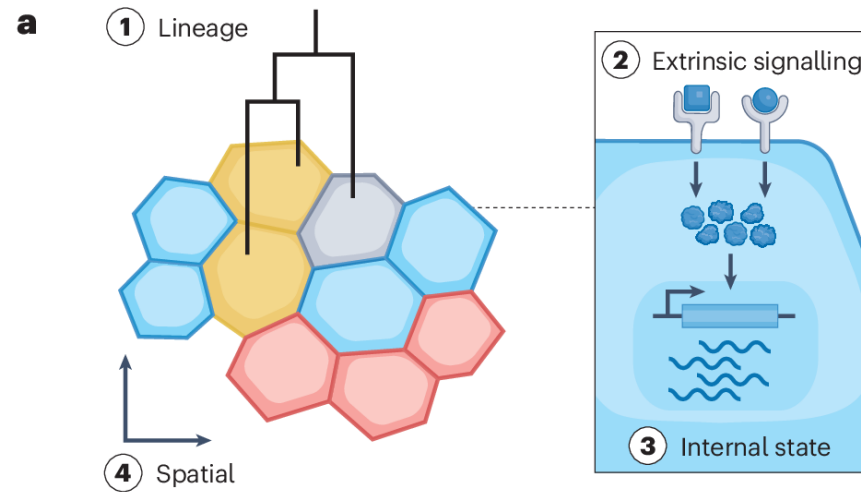
Open directions



Jones MG, et al. 2023
Annu. Rev. Cancer Biol. 7:311–29

Increasing adoption in other areas of biology, e.g. cancer & immunology

Open directions

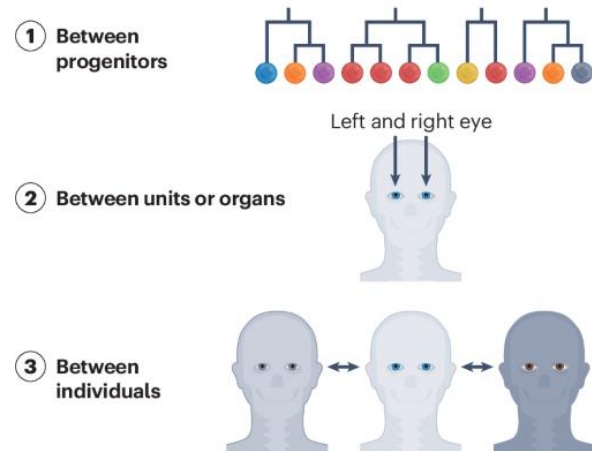


Increasing adoption in other areas of biology, e.g. cancer & immunology

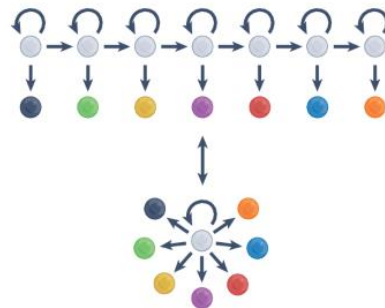
Linking molecular modalities

Open directions

Analyse variability



Quantify and compare statistical rules



Increasing adoption in other areas of biology, e.g. cancer & immunology

Linking molecular modalities

Statistical challenges

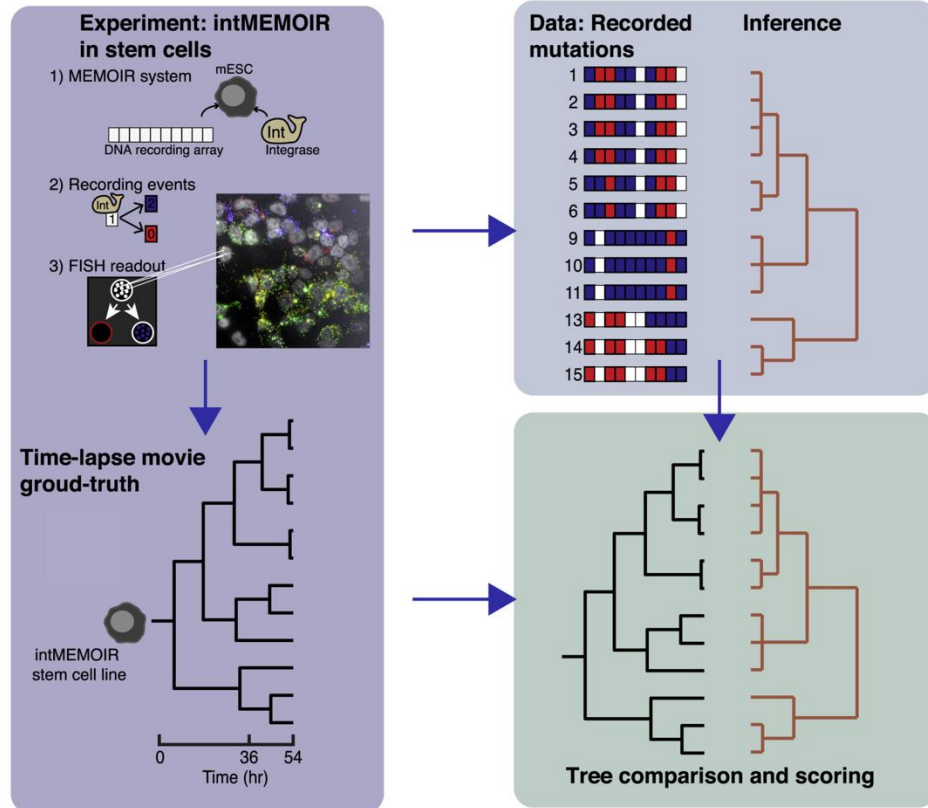
Acknowledgements

Louis du Plessis

Antoine Zwaans

Laura Thomas

TiDeTree Tutorial



- 10 (or 3 alignments) of 100 alignments
- intMEMOIR system
- Recombinase

Data

TiDeTree Tutorial

Reconstructing time-scaled single-cell phylogenies from genetic lineage tracing data
by Sophie Seidel

Tutorial

-  Github repository
-  License
-  Statistics

Data

-  alignment_1.tidetree
-  alignment_10.tidetree
-  alignment_2.tidetree
-  alignment_3.tidetree
-  alignment_4.tidetree
-  alignment_5.tidetree
-  alignment_6.tidetree
-  alignment_7.tidetree
-  alignment_8.tidetree
-  alignment_9.tidetree

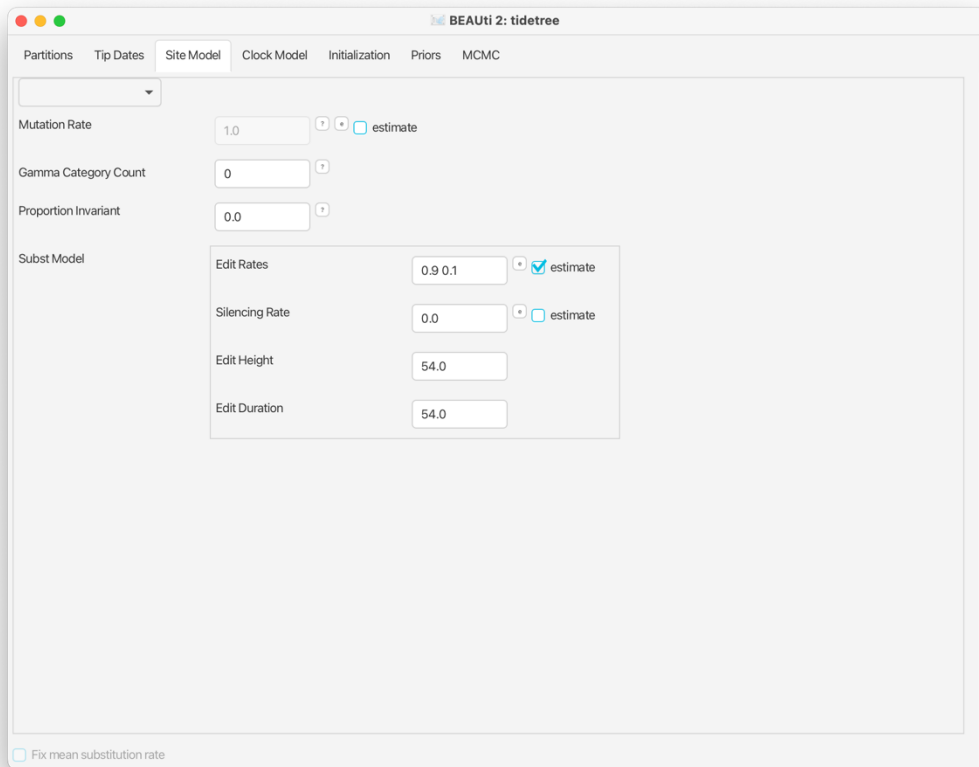
Background

Understanding how cells divide, differentiate, and die over time is central to developmental biology and cancer research. Recent advances in single-cell lineage recording allow us to track cell histories—but inferring developmental dynamics from these data requires statistical tools ([Askary et al., 2024](#)).

TiDeTree ([Seidel & Stadler, 2022](#)) is a BEAST2 package designed for statistical inference from such single-cell recording data. It jointly infers time-scaled single-cell phylogenies and editing model parameters, including editing rates (analogous to molecular clock rates) and the probabilities of different editing outcomes. Beyond tree reconstruction, TiDeTree enables the inference of cell population dynamics, such as cell division, death and differentiation rates.

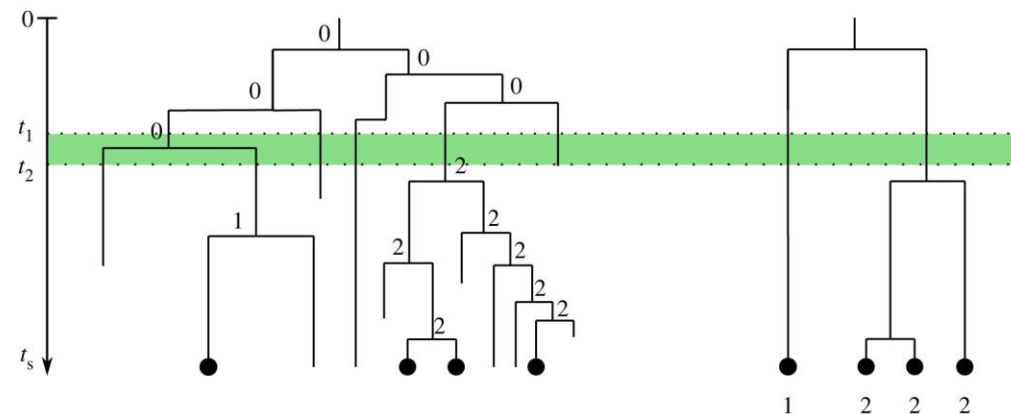
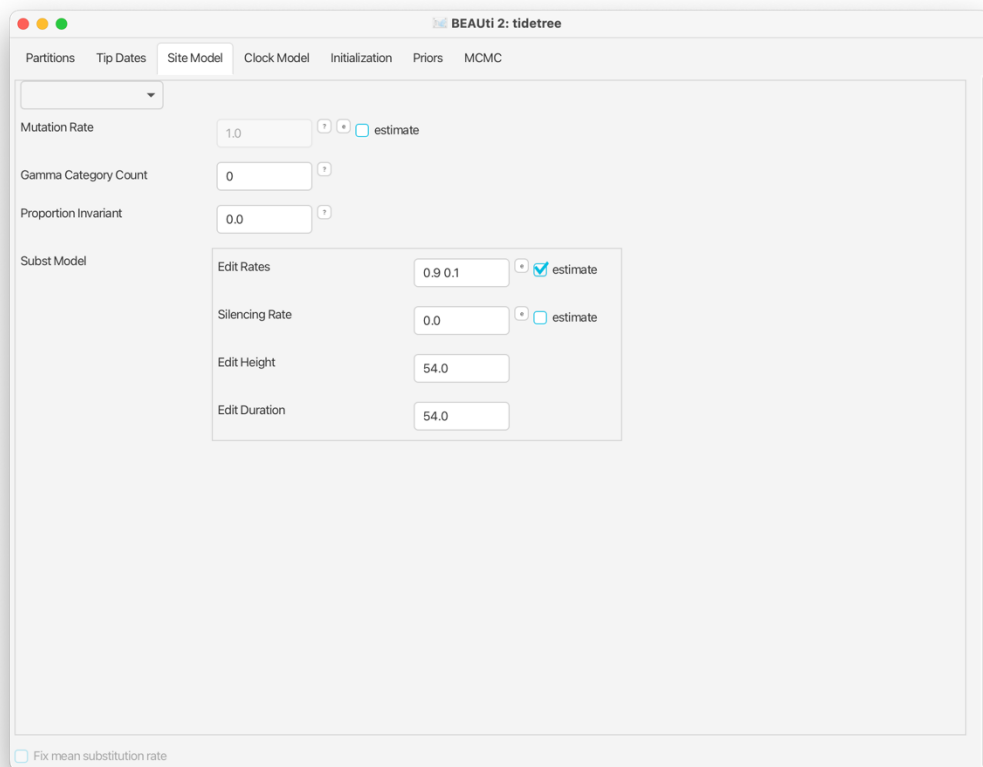
This tutorial will guide you through the setup and application of TiDeTree using an example dataset. You will learn how to model the editing process, reconstruct timed cell phylogenies, and estimate parameters describing the underlying cellular dynamics.

TiDeTree likelihood with piecewise constant editing



$$Q_{(t_1 \leq t \leq t_2)} = \begin{pmatrix} -l - r \times \sum_i s_i & l & r \times s_1 & \dots & r \times s_S \\ 0 & 0 & 0 & \dots & 0 \\ 0 & l & -l & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & l & 0 & \dots & -l \end{pmatrix}$$

TiDeTree likelihood with piecewise constant editing



Feedback time!

